

Brevity Is Not All You Need: A Stylometric and Structural Case Study of Zero-Shot LLM Code Synthesis vs. Production-Hardened Human Reference Code

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Abstract

Evaluating Large Language Model (LLM) code generation typically focuses on functional pass rates (e.g., HumanEval, MBPP) rather than stylometric, performance, and maintenance properties. This paper presents an empirical case study comparing **147 zero-shot synthetic code generations** produced by three frontier LLM architectures (*Google Gemini 3.5 Flash*, *OpenAI GPT-5.6 Sol*, and *Anthropic Claude Sonnet 4.6*) across 14 algorithmic tasks generated via isolated agent sessions against a reference baseline of 10 production-hardened standard library functions authored by senior human engineers prior to the LLM era (2017–2018 code from React 16, Go 1.10, Redis 5.0, Linux Kernel 4.14, Rust stdlib, PyTorch 1.0, and FastHTTP). We also evaluate an auxiliary secondary benchmark of 50 pilot recreation generations produced by Gemini 3.5 Flash across the 10 human prompt tasks (total master dataset N=207).

Quantitative analysis demonstrates statistically significant stylometric divergence: frontier synthetic implementations exhibit **+221% lines of code (LOC) expansion** (Mean = 48.13 ± 26.36 LOC vs. 15.00 ± 6.78 LOC human, Mann-Whitney U = 102.5, Holm-Bonferroni $p_{adj} = 3.32 \times 10^{-5}$, rank-biserial effect size $r_{rb} = +0.861$), elevated comment density ($9.56\% \pm 10.82\%$ synthetic vs. $1.43\% \pm 4.52\%$ human, $p_{adj} = 0.0018$, $r_{rb} = +0.621$), higher explicit type annotation density (8.12 ± 8.95 vs 1.50 ± 1.35 , $p_{adj} = 0.0003$, $r_{rb} = +0.712$), and higher vertical whitespace ratios ($15.92\% \pm 4.88\%$ synthetic vs. $5.54\% \pm 6.95\%$ human, $p_{adj} = 0.0002$, $r_{rb} = +0.765$). All confidence intervals are computed using non-parametric 10,000-resample bootstrapping.

Glossary of Technical & Statistical Terms for General Readers

- **Lines of Code (LOC):** The total count of executable and structural code lines, excluding blank lines.
- **Mann-Whitney U Test:** A non-parametric statistical test that compares two independent groups without assuming a bell-curve distribution.
- **Holm-Bonferroni FWER Correction (p_{adj}):** A statistical procedure adjusting p-values to prevent false positive discoveries during multiple comparisons.
- **Rank-Biserial Correlation (r_{rb}):** An effect size metric (-1.0 to +1.0) indicating how strongly one group's values exceed another.
- **Bootstrap 95% Confidence Interval:** A computational resampling method estimating uncertainty bounds by repeatedly sampling data 10,000 times.
- **Kruskal-Wallis H-Test:** A statistical test evaluating whether three or more independent model families differ significantly.
- **Structural Micro-Fragmentation:** The tendency of synthetic models to decompose simple logic into multiple helper functions or extra class wrappers.

5. Quantitative Stylometric Results & Statistical Significance

5.1 Primary Frontier Model Study: Human Reference (n=10) vs. Frontier LLMs (N=147)

Stylometric Metric	Human Reference (n=10)	Frontier LLMs (N=147)	Mann-Whitney U	Raw p-val	Holm-Bonferroni p_{adj}	Rank-Biserial Effect Size (r_{rb})	FWER Significance
Lines of Code (LOC)	15.00 ± 6.78 [11.1, 18.9]	48.13 ± 26.36 [43.8, 52.5]	102.5	p = 5.53e-06	$p_{adj} = 3.32e-05$	$r_{rb} = +0.861$ (Massive)	Significant (p < 0.01)
Comment Density (%)	1.43% ± 4.52% [0.0, 4.3]	9.56% ± 10.82% [7.8, 11.3]	280.0	p = 3.12e-04	$p_{adj} = 0.0018$	$r_{rb} = +0.621$ (Large)	Significant (p < 0.01)
Explicit Type Annotations	1.50 ± 1.35 [0.7, 2.3]	8.12 ± 8.95 [6.6, 9.6]	212.0	p = 6.15e-05	$p_{adj} = 0.0003$	$r_{rb} = +0.712$ (Large)	Significant (p < 0.01)
Vertical Whitespace (%)	5.54% ± 6.95% [1.8, 9.8]	15.92% ± 4.88% [15.1, 16.7]	172.0	p = 3.10e-05	$p_{adj} = 0.0002$	$r_{rb} = +0.765$ (Massive)	Significant (p < 0.01)



5.2 Per-Model Inter-Model Significance Tests (N=147)

Model Sub-Group	Record Count (N)	Mean LOC (± SD)	Mean Comment Density (%)
Google Gemini 3.5 Flash	N=81	47.74 ± 22.81	14.02% ± 12.11%
OpenAI GPT-5.6 Sol	N=33	39.24 ± 32.83	0.45% ± 1.61%
Anthropic Claude Sonnet 4.6	N=33	57.97 ± 24.67	7.74% ± 10.10%
Kruskal-Wallis H-test	—	H = 15.38, $p_{adj} = 4.58e-04$	H = 41.35, $p_{adj} = 2.08e-09$

6. Conclusion & References

This study establishes statistically significant stylometric differences between zero-shot LLM code and production-hardened human reference code across 207 total program records, proving that LOC bloat is prompt-scope dependent and model-family specific.